Technische Hochschule Deggendorf

Case Study Project – FactoryDocs Assistant: A Retrieval-Augmented Chatbot for Robot Error Diagnosis in Production

Bibliography Report

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Abstract

Modern production increasingly relies on collaborative robots (cobots) that load machines, move parts and work next to people. When such a robot stops itself with a protective stop, the connected machine and the material flow stop as well, and the manufacturer requires the root cause to be reviewed before the robot may restart. In practice this review is slow: the explanation of each error code, the robot settings, gripper problems and repair steps are spread over several long manuals, operators describe problems in their own words, and general-purpose AI chatbots answer from memory without a verifiable source.

This report presents a bibliographic study of the methods needed to solve this problem with Retrieval-Augmented Generation (RAG). It moves from the general to the specific: from Industry 4.0 and the cost of downtime, through technical language processing and Transformer-based language models, to keyword, dense and hybrid retrieval, RAG and its evaluation, and finally to recent studies that apply RAG to industrial troubleshooting manuals. Twenty-six research works, together with the relevant safety standard and manufacturer guidance, are summarized by their method, main result and limitation, and are compared in a summary table.

The review shows that RAG grounds answers in documents and makes them traceable, which industrial studies identify as essential for operator trust; that keyword search remains strong for exact terms such as error codes while dense retrieval handles paraphrased descriptions, which motivates hybrid retrieval; and that RAG must be evaluated on retrieval, answer correctness and the ability to refuse. Based on these findings, the report justifies the method of the FactoryDocs Assistant: a local, hybrid RAG chatbot over four public Universal Robots UR5e and Robotiq gripper manuals, evaluated on a 60-question test set against plain-LLM, BM25, dense and hybrid configurations.

Keywords: Retrieval-Augmented Generation, robot error diagnosis, collaborative robots, technical language processing, hybrid retrieval, local large language models, industrial maintenance.

1. Introduction

The industrial shift toward Industry 4.0 connects machines, sensors and software into cyber-physical systems that monitor themselves and support human decisions [1]. Industrial robots are a central part of this shift: about 542,000 industrial robots were installed worldwide in 2024, and the operational stock reached about 4.66 million units [2]. Collaborative robots such as the Universal Robots UR5e are designed to work close to people and are used for machine tending, pick-and-place and assembly.

As more production steps depend on robots, the cost of a stopped robot grows. A robot that loads a CNC machine is part of a chain: when it stops, the machine waits, the buffers fill or empty, and the material flow to and from the warehouse is interrupted. A 2024 Siemens study estimates that unplanned downtime costs the world's 500 largest companies about $1.4 trillion per year, around 11% of their revenue [3]. This chapter introduces the problem, the objectives and the scope of this report.

1.1 Problem Statement

Cobots stop themselves for safety. A protective stop is triggered when the robot detects, for example, a collision or a deviation from its planned path, and the controller shows an error code such as C153 (position deviates from path) or C157 (collision detected by joint). Universal Robots lists a wrong payload or mounting, external cables dragging on the robot, collisions, start-up in cold environments and very high accelerations combined with blends as common causes; it prohibits the automatic reset of protective stops and warns that resetting them without fixing the root cause is misuse [4]. Its service note adds that a protective stop shall not be reset without a review to determine the fault's root cause [5]. In practice, this review is slow for three reasons:

- **The knowledge is scattered.** The meaning of a code, robot settings, gripper problems and repair steps are found in different documents: an error code directory, a user manual, a gripper manual and a service handbook.
- **Workers use their own words.** An operator may type "the robot keeps dropping parts" instead of the exact term in the manual, so keyword search (Ctrl+F) often fails.
- **General chatbots are not reliable enough.** Large language models (LLMs) answer fluently from memory but can invent plausible, wrong details and give no source, which is unacceptable when the answer affects machine safety.

1.2 Objectives

- To review the literature from the general industrial context to the specific methods of the project: language models, retrieval, RAG and its evaluation.
- To summarize the method, results and limitation of each reviewed work in our own words.
- To identify the research gaps that the project addresses.

- To justify the materials and methods of the FactoryDocs Assistant based on the literature.

1.3 Scope of Study

This review covers peer-reviewed journal articles, conference papers, widely cited preprints and official industry documents published between 2009 and 2026, with an emphasis on retrieval and RAG for technical and industrial documents. Foundational works on retrieval and Transformers are included because the project builds directly on them. Image-based (multimodal) document understanding and model fine-tuning are outside the scope of this report.

2. Materials and Methods

This chapter describes how the literature was searched, selected and analysed, and which materials the project uses.

2.1 Literature Search Method

Sources were searched in IEEE Xplore, the ACM Digital Library, arXiv, the proceedings of the PHM Society and Google Scholar. The search combined three groups of terms:

- **Method terms:** retrieval-augmented generation, dense retrieval, hybrid search, BM25, hallucination, large language model.
- **Domain terms:** technical manual, troubleshooting, fault diagnosis, maintenance, industrial robot.
- **Evaluation terms:** RAG evaluation, faithfulness, negative rejection.

The search was carried out between 19 and 30 October 2026 and was limited to works in English. Records were first screened by title and abstract and then read in full; the exact number of records at each step was not recorded. Citation searching (backward snowballing) in the most relevant papers added further works: the reference list of Rossi et al. [25] led to the studies by Alfeo et al. [8], Kiangala and Wang [28] and Ren et al. [27], and the reference list of Lukens et al. [26] led to the study by Löwhagen et al. [29]. All of these works were then read in their original form.

2.2 Selection Criteria

A work was included when it was published in a peer-reviewed journal or conference, or was a widely cited preprint, and met at least one of three rules:

- **Method rule:** it defines a method used in the project, such as BM25, dense retrieval or RAG.
- **Evaluation rule:** it evaluates retrieval or RAG on technical or industrial documents, or defines how RAG should be evaluated.
- **Context rule:** it is a reliable source on the industrial problem, such as a robot safety standard, manufacturer documentation or an industry study.

2.3 Data Extraction and Analysis

For every included work, the domain, data, method, main quantitative result and main limitation were recorded. The works were grouped into six topics that move from the general to the specific: (1) industrial context and maintenance knowledge, (2) language models and hallucination, (3) retrieval, (4) RAG, (5) RAG evaluation and (6) RAG for industrial troubleshooting. Each summary in Chapter 3 states what the work does, its result and its limitation, followed by a key insight and its relevance to this project.

2.4 Selection of Input Documents

The input documents form the knowledge base from which the assistant answers. Because every answer must be traceable to an authoritative source, the documents were selected with the same care as the literature. A document was included only if it met all six criteria below.

1. **Relevance to the use case:** it covers the UR5e robot or its Robotiq gripper in a machine-tending cell.
2. **Authoritative source:** it is official documentation published by the manufacturer, not a forum post, blog or video.
3. **Public and legal access:** it can be downloaded free of charge, so that others can reproduce the project; the files are linked, not redistributed.
4. **Matching version:** it applies to the e-Series hardware and the PolyScope 5 software used in the use case.
5. **Machine-readable:** the PDF has a text layer, so text and tables can be extracted without optical character recognition.
6. **Coverage of the question types:** together, the documents answer all five question types of the test set (Table 2.3).

Excluded were unofficial sources such as forums and videos, manuals for the older CB-Series robots and for the PolyScope X software, and documents for other robot models or grippers. Table 2.1 shows which document is expected to answer which type of question.

Table 2.1: Expected coverage of question types by the input documents.

Document	**Question types it supports**
UR Error Codes Directory	Error code: meaning and suggested fix of codes such as C153 or C157
UR5e User Manual	Problem description and procedure: payload, installation, safety settings and robot stops
Robotiq 2F-85/2F-140 Manual	Problem description: dropped or lost parts (object detection and re-grasp, Section 4.5); gripper fault and communication troubleshooting (Chapter 8)
UR e-Series Service Handbook	Procedure: repair and troubleshooting steps
All four together	Multi-document questions; unanswerable questions are those that none of them covers

2.5 Project Materials

The use case is a Universal Robots UR5e cobot that tends a CNC machine in a production cell, with automated guided vehicles (AGVs) supplying and removing parts. When the robot triggers a protective stop, the operator types the error code or describes the problem, and the assistant returns the meaning of the error, likely causes, fix steps and source pages.

The knowledge base consists of the four selected manuals (Table 2.2). All four PDFs were checked before use: each has a text layer, so no optical character recognition is needed. The three Universal Robots documents belong to PolyScope 5, with software versions 5.24 to 5.26; the small version differences are recorded so that answers can be traced to the exact edition. The Error Codes Directory was also checked against the use case: it lists C153 (position deviates from path) and C157 (collision detected by joint) on pp. 34–35, and for code C157A2 (detected by the elbow joint) it suggests making sure no objects are in the path and checking the payload, centre of gravity and acceleration settings [32]. Both codes appear in the directory's chapter for CB3 and earlier models, but Universal Robots lists them as the common protective-stop codes for its current robots as well [4]. The Service Handbook states that service and troubleshooting work should only be performed by qualified personnel [35], which the assistant must respect. The documents are not stored in the project repository because of copyright; the repository provides download links only.

Table 2.2: Documents in the knowledge base.

Document	**Version**	**Pages**	**Publisher**
UR Error Codes Directory [32]	PolyScope 5, SW 5.26, doc. version 10.19.440 (2026)	174	Universal Robots
UR5e User Manual [33]	PolyScope 5, SW 5.24, doc. version 10.15.254 (2025)	237	Universal Robots
Robotiq 2F-85/2F-140 Manual for UR [34]	Revision 2020/02/11 (e-Series and CB-Series)	153	Robotiq
UR e-Series Service Handbook [35]	SW 5.25, doc. version 10.16.265 (2025)	126	Universal Robots

The test set consists of 60 questions with reference answers and source pages, written and checked by all three team members (Table 2.3). Unanswerable questions are checked against all four manuals to make sure that none of them contains the answer.

Table 2.3: Question types in the test set.

Question type	**Example**	**Planned**
Error code	"What does C157A2 mean and how do I fix it?"	15
Problem description	"The robot keeps dropping parts."	15
Multi-document	"The robot stops after a heavier part was introduced. What should I check?"	10
Procedure	"How do I set the payload for a new gripper?"	10
Unanswerable	"How long does delivery of a replacement joint take?"	10

3. Bibliography Summary and Literature Alignment

3.1 Overview

This chapter summarizes the research works that shaped the design of the FactoryDocs Assistant. It starts with the industrial background and the nature of maintenance knowledge, then covers the language models and retrieval methods that make document-based answering possible, and ends with studies that apply these methods to industrial troubleshooting manuals.

3.2 Industrial Context and Maintenance Knowledge

3.2.1 A Cyber-Physical Systems Architecture for Industry 4.0

Authors: J. Lee, B. Bagheri and H.-A. Kao (2015), Manufacturing Letters [1]

This paper proposes the 5C architecture for cyber-physical production systems: connection, conversion, cyber, cognition and configuration. Machine data is collected, turned into information, compared across machines, presented to people for decisions at the cognition level, and finally used to adjust the system. The paper is conceptual and does not evaluate an implementation.
Key Insight: Data creates value only when the right information reaches decision-makers at the right time.
Relevance: The FactoryDocs Assistant works at the cognition level, giving operators the right manual information during a robot stop.

3.2.2 Collaborative Robot Safety and Protective Stops

Authors: ISO/TS 15066 (2016) [6]; Universal Robots (2020–2026) [4, 5, 33]

ISO/TS 15066 specifies safety requirements for collaborative operation, including limits on the forces a robot may exert on a person; cobots therefore monitor forces and joint torques and stop themselves when a limit is exceeded. Universal Robots lists a wrong payload or mounting, dragging external cables, collisions, cold start-up and very high accelerations as common causes of protective stops, and prohibits their automatic reset [4]; its service note states that a stop shall not be reset without a review of its root cause [5]. The current UR5e user manual notes that this event is now called a “robot stop” and was previously known as a “protective stop” (PDF p. 216) [33]. None of these sources provides tools for performing the root-cause review quickly.
Key Insight: Protective stops are a built-in safety function, and each one requires a root-cause review.
Relevance: This defines the exact task the assistant supports: fast, documented root-cause diagnosis.

3.2.3 Technical Language Processing

Authors: M. P. Brundage, T. Sexton, M. Hodkiewicz, A. Dima and S. Lukens (2021), Manufacturing Letters [7]

The authors argue that out-of-the-box NLP pipelines, designed for non-technical language, fail on maintenance text such as work orders, which is short, messy and full of abbreviations, codes and asset numbers. Using real work-order examples, they show typical failures: a tokenizer does not split "hydpump", standard stop-word removal deletes "not" from a record such as "hydpump not working" and reverses its meaning, cleaning removes asset numbers such as HS012, and stemming fails to merge "hyd", "hydraulic" and the misspelling "hydaulic". They propose technical language processing (TLP), a human-in-the-loop, iterative approach that tailors NLP tools to engineering data together with domain experts. The paper is a short position letter with illustrative examples and reports no new experiments.
Key Insight: Technical text needs methods that handle exact codes as well as informal descriptions.
Relevance: Our test set mixes exact error codes ("C157A2") with free-text problems ("the robot keeps dropping parts"), and our pipeline does not remove stop words before retrieval.

3.2.4 Troubleshooting with Sentence Embeddings

Authors: A. L. Alfeo, M. G. C. A. Cimino and G. Vaglini (2021), Journal of Intelligent Manufacturing [8]

This study retrieves solutions from 308 problem–solution pairs in the technical assistance reports of a machinery manufacturer. Problem descriptions are cleaned (punctuation and stop-word removal, lemmatization), embedded with a pre-trained Sentence-BERT model that is fine-tuned on 2,440 expert-rated problem pairs (after synonym-based augmentation), and ranked by their cosine similarity, refined by a small neural scoring network. On a relevance set labelled by eight technicians, it reached a MAP@5 of 0.71–0.76 and an MRR@5 of 0.75–0.82, clearly ahead of BM25, bag-of-words cosine similarity and the Universal Sentence Encoder. The domain knowledge was collected with short (about 30-minute) technician questionnaires. The approach needs a history of solved cases, which a new robot cell may not have, and its data set is small and not public.
Key Insight: Meaning-based retrieval can find relevant troubleshooting knowledge that keyword search misses, with little expert effort.
Relevance: Supports the dense retrieval component of our system and shows that BM25 alone can be weak on informal technical descriptions.

3.3 Language Models and Hallucination

3.3.1 Attention Is All You Need

Authors: A. Vaswani et al. (2017), Advances in Neural Information Processing Systems [9]

This paper introduces the Transformer, an architecture based entirely on attention, in which every word weighs its relationship to every other word and training runs in parallel. On the WMT 2014 benchmarks it reached 28.4 BLEU for English–German and

41.0 BLEU for English–French at a fraction of the training cost of earlier models. The proceedings version evaluates machine translation only.
Key Insight: Attention captures context efficiently and became the basis of nearly all modern language models.
Relevance: Both the embedding model and the LLM in our pipeline are Transformers.

3.3.2 BERT

Authors: J. Devlin, M.-W. Chang, K. Lee and K. Toutanova (2019), NAACL-HLT [10]

BERT is a Transformer encoder pre-trained by predicting masked words and then fine-tuned for each task. It set new results on eleven NLP tasks, raising the GLUE score to 80.5 and SQuAD v1.1 F1 to 93.2. As Reimers and Gurevych later showed, comparing two sentences with BERT requires passing both through the network together, which is too slow for large-scale search [15].
Key Insight: Pre-trained encoders produce strong, context-aware text representations.
Relevance: BERT is the basis of the sentence-embedding models used for semantic search.

3.3.3 The Llama 3 Herd of Models

Authors: A. Grattafiori et al. (Llama Team, AI @ Meta) (2024), arXiv [11]

This report describes a family of LLMs with 8, 70 and 405 billion parameters, pre-trained on about 15 trillion tokens and supporting contexts of up to 128,000 tokens; all reported results refer to the Llama 3.1 release, whose weights are publicly available. The report evaluates general benchmarks, not technical manuals, and the small model is less capable than the largest ones. Running the 8B model locally through Ollama is a design choice of this project, not a result of the report.
Key Insight: Capable open-weight models can now run locally without sending data to the cloud.
Relevance: A local Llama 3.1 8B model generates the answers, keeping company documents on site.

3.3.4 Survey of Hallucination in Natural Language Generation

Authors: Z. Ji et al. (2023), ACM Computing Surveys [12]

This survey defines intrinsic hallucination (output that contradicts the source) and extrinsic hallucination (output that cannot be verified from the source), and reviews causes, metrics and mitigation across summarization, dialogue, generative question answering, data-to-text generation and machine translation. Augmenting the input with retrieved external knowledge is one of the mitigation strategies it describes. As a survey it reports no new experiments.
Key Insight: An LLM used alone cannot be trusted for exact technical values.
Relevance: Motivates the plain-LLM baseline and the "I don't know" requirement.

3.4 Keyword, Dense and Hybrid Retrieval

3.4.1 The Probabilistic Relevance Framework: BM25

Authors: S. Robertson and H. Zaragoza (2009), Foundations and Trends in Information Retrieval [13]

This work explains the framework behind BM25, which scores a document higher when it contains the query terms often, when those terms are rare in the collection, and when the document is not unusually long. BM25 needs no training data and is fast and interpretable. It only matches words that literally appear in both query and document.
Key Insight: Keyword ranking is a strong, cheap baseline for exact terms.
Relevance: BM25 is our keyword baseline and part of the hybrid retriever.

3.4.2 Dense Passage Retrieval

Authors: V. Karpukhin et al. (2020), EMNLP [14]

Questions and passages are encoded by two BERT-based encoders and ranked by vector similarity over about 21 million Wikipedia passages. On open-domain question answering, DPR outperformed a strong BM25 system by 9 to 19 percentage points in top-20 retrieval accuracy. The encoders need training pairs, and the evaluation covers general knowledge only.
Key Insight: Dense retrieval finds relevant passages even when they use different words.
Relevance: Supports dense retrieval for problems described in the operator's own words.

3.4.3 Sentence-BERT

Authors: N. Reimers and I. Gurevych (2019), EMNLP-IJCNLP [15]

This paper fine-tunes BERT in siamese and triplet networks so that each sentence maps to one comparable vector. Finding the most similar pair among 10,000 sentences drops from about 65 hours with BERT to about 5 seconds, while keeping competitive accuracy on semantic similarity tasks. The models are trained on general-domain data.
Key Insight: Sentence embeddings make semantic search practical.
Relevance: Our chunks are embedded with a sentence-transformer model.

3.4.4 MiniLM

Authors: W. Wang, F. Wei, L. Dong, H. Bao, N. Yang and M. Zhou (2020), Advances in Neural Information Processing Systems [16]

MiniLM compresses a large Transformer into a small student by distilling its self-attention behaviour. The student keeps more than 99% of the teacher's accuracy on SQuAD 2.0 and several GLUE tasks with half of its parameters and computation. Some accuracy is lost compared with larger encoders.
Key Insight: Small distilled encoders keep most of the quality at a fraction of the cost.
Relevance: The all-MiniLM-L6-v2 model, based on this work, runs efficiently on a laptop.

3.4.5 FAISS: Billion-Scale Similarity Search with GPUs

Authors: J. Johnson, M. Douze and H. Jégou (2021), IEEE Transactions on Big Data [17]

This paper presents GPU algorithms for exact and approximate nearest-neighbour search, released as the open-source Faiss library. It reports search 8.5 times faster than the previous GPU state of the art and builds a k-nearest-neighbour graph connecting one billion vectors in under 12 hours on four GPUs. Approximate search trades some accuracy for speed.
Key Insight: Vector search can be made fast enough for interactive use.
Relevance: FAISS stores and searches the embedded manual chunks.

3.4.6 BEIR: A Benchmark for Zero-Shot Retrieval

Authors: N. Thakur, N. Reimers, A. Rücklé, A. Srivastava and I. Gurevych (2021), NeurIPS Datasets and Benchmarks Track [18]

BEIR evaluates ten retrieval systems, including lexical, sparse, dense, late-interaction and re-ranking models, on 18 datasets from different domains without domain-specific training. BM25 proved a robust baseline, and several dense retrievers trained on one domain fell behind it on others, while re-ranking models performed best on average at higher cost. The benchmark contains no industrial manuals.
Key Insight: Neither keyword nor dense retrieval wins in every domain.
Relevance: Justifies combining BM25 and dense search for a new domain such as robot manuals.

3.4.7 Reciprocal Rank Fusion

Authors: G. V. Cormack, C. L. A. Clarke and S. Buettcher (2009), ACM SIGIR [19]

RRF combines several ranked lists by giving each document a score of $1/(k + \text{rank})$ in every list, with $k = 60$, and summing the scores. Across pilot runs and four TREC tasks, RRF beat the Condorcet Fuse method in every case and the CombMNZ method in all but one, and it beat the best individual run except where that run involved a human in the loop; on average it was 4–5% better. On the LETOR 3 dataset, fusing several learning-to-rank methods with RRF outperformed each of them. The authors found $k = 60$ near-optimal but not critical. RRF uses ranks only, so one retriever cannot be weighted more than another without changing the method.
Key Insight: A simple fusion rule reliably combines different retrievers.
Relevance: RRF fuses the BM25 and dense rankings in our hybrid retriever.

3.5 Retrieval-Augmented Generation

3.5.1 Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks

Authors: P. Lewis et al. (2020), Advances in Neural Information Processing Systems [20]

A DPR retriever selects passages from a Wikipedia index, and a BART generator writes the answer conditioned on them, combining learned (parametric) memory with documents (non-parametric memory). RAG set new results on three open-domain question answering tasks and produced more specific and factual text than a generator alone; the index can be replaced without retraining. Retrieval matters: replacing the learned retriever with BM25 or freezing it lowered results on most tasks, and BM25 was best only on the entity-centric FEVER task.
Key Insight: Grounding generation in retrieved documents makes answers more factual and updatable.
Relevance: RAG is the core architecture of the FactoryDocs Assistant.

3.5.2 Retrieval-Augmented Generation for Large Language Models: A Survey

Authors: Y. Gao et al. (2023), arXiv [21]

This survey classifies RAG systems as naive, advanced and modular, and describes the main design choices: chunking, retrieval including hybrid search and re-ranking, prompt augmentation and generation. It lists open problems such as robustness to irrelevant context and evaluation. As a survey it gives no new measurements.
Key Insight: RAG quality depends on many design choices, not only on the LLM.
Relevance: Guides our chunking, retrieval and prompt design.

3.5.3 Lost in the Middle

Authors: N. F. Liu et al. (2024), Transactions of the Association for Computational Linguistics [22]

The authors move the relevant document to different positions in the input for multi-document question answering and key–value retrieval. Accuracy follows a U-shaped curve: it is highest when the relevant information is at the beginning or end and drops when it is in the middle. With the relevant document in the middle, GPT-3.5-Turbo even scored lower than without any documents (closed-book, 56.1%). Adding more retrieved text therefore does not guarantee better answers. The study uses general-domain data and the models available in 2023.
Key Insight: More context is not always better.
Relevance: We keep top-k small and test different top-k values.

3.6 Evaluating RAG Systems

3.6.1 RAGAS: Automated Evaluation of RAG

Authors: S. Es, J. James, L. Espinosa-Anke and S. Schockaert (2024), EACL System Demonstrations [23]

RAGAS scores RAG pipelines without reference answers by prompting an LLM (gpt-3.5-turbo-16k in the paper) for three metrics: faithfulness (the share of claims in the answer that the retrieved context supports), answer relevance and context relevance. The authors built WikiEval, 50 Wikipedia pages on events from 2022 onwards, annotated by two people. RAGAS agreed with the human judgments in 95% of pairwise comparisons for

faithfulness, 78% for answer relevance and 70% for context relevance, clearly better than asking ChatGPT directly for a score (72%, 52% and 63%). The authors found context relevance the hardest aspect to judge, because the LLM struggled to select the crucial sentences from longer contexts; all scores were produced with a single judge model.
Key Insight: Faithfulness can be measured automatically, but needs a human cross-check.
Relevance: RAGAS complements our two-person rubric.

3.6.2 Benchmarking LLMs in Retrieval-Augmented Generation

Authors: J. Chen, H. Lin, X. Han and L. Sun (2024), AAAI Conference on Artificial Intelligence [24]

The RGB benchmark tests four abilities of LLMs in RAG: noise robustness, negative rejection (refusing when the documents lack the answer), information integration (combining facts from several documents) and counterfactual robustness, using 600 base questions plus 200 for integration and 200 for counterfactual robustness, half in English and half in Chinese. Six LLMs were tested. Accuracy fell as noise grew, for ChatGPT from 96.33% to 76.00% at 80% noisy documents. When all documents were irrelevant, the best rejection rates were only 45% in English and 43.33% in Chinese, and on integration questions the best accuracy without noise was only 60% and 67%. The questions are built from general news, not technical documents.
Key Insight: Refusing correctly and combining several sources are known weak points.
Relevance: Motivates our refusal KPI and the multi-document and unanswerable question types.

3.7 RAG and LLMs for Industrial Troubleshooting

3.7.1 RAG for Troubleshooting a Large Cyber-Physical System

Authors: M. T. Rossi, L. Mariani, O. Riganelli, G. Filomeno, D. Giannone and P. Gavazzo (2026), ICSE-SEIP [25]

The authors study a fully local RAG system, run with Ollama, for troubleshooting procedures from more than 100 manuals of a naval system at Fincantieri. They selected 25 sets of failure symptoms with their procedures and wrote four questions for each (accurate or imprecise wording, with or without context), 100 questions in total, and tested three local models (Llama 3B, Mistral 7B, Qwen 7B) with chunk sizes of 400, 800 and 1,000 characters, giving 135,000 assessed answers. The correct block appeared among the top four retrieved blocks for 41% of accurately worded questions but only 19% of imprecise ones; still, in a manual inspection of the best configuration, 64% of answers to accurate questions with context contained all troubleshooting steps and only 8% were wrong, rising to 56% wrong for imprecise questions without context. Average response times ranged from 3.6 s (3B model) to 12.2 s (7B model). The authors conclude that answers must show their sources so that experts can validate and complete them. Limitations include LLM-based scoring, only 25 topics and no involvement of practitioners in the assessment.
Key Insight: RAG helps operators when every answer shows its source, but wording strongly affects retrieval.
Relevance: Supports local deployment, source citations, mixed-wording questions and our latency target.

3.7.2 Evaluation Framework for Fault Diagnosis Using Technical Manuals

Authors: S. Lukens, M. Bishof, N. Siddiqui and D. West (2025), Annual Conference of the PHM Society [26]

The authors use a public US Army operator manual for light medium tactical vehicles, listing 224 malfunctions in 22 systems, and 300 operator observations generated synthetically with GPT-4.1; the system itself uses GPT-3.5-Turbo, PyMuPDF for PDF extraction, the all-MiniLM-L6-v2 encoder and 1,000-character chunks. They compare an LLM alone, three RAG variants and a retrieval-only baseline that matches observations to the fault names with the same MiniLM embeddings, without an LLM. The fault codes are named symptoms such as "ENGINE OVERHEATS" rather than alphanumeric codes. In a chat-style task, the hallucination rate, measured by matching suggestions against the manual's fault list, fell from 94.4% for the LLM alone to 77–79% with RAG. In a label-constrained task, retrieval alone reached an 86.3% hit rate against 69.6% for the LLM alone and 57.3–64.3% for RAG, and was by far the fastest (117 s versus 301–334 s per 100 observations). The authors stress that the synthetic data set is narrow and that the public manual likely appeared in the LLM's training data, so proprietary manuals may behave differently.
Key Insight: For matching observations to a fixed list of fault names, embedding retrieval alone beat both the LLM and RAG; systems must be evaluated before they are demonstrated.
Relevance: Supports our retrieval and plain-LLM baselines and our evaluation-first approach.

3.7.3 Voice Query for Industrial Robot Maintenance

Authors: Z. Ren, Z. Yu, W. Zhang and Q. Lei (2024), IEEE Access [27]

Ren et al. build a voice-driven assistant for industrial-robot fault diagnosis. Speech is transcribed with Whisper; the robot user manual is split into chunks (size 500, overlap 25), embedded with OpenAI's text-embedding-ada-002 and stored in a clustered vector database; the ChatGLM3-6B model generates the answer from the retrieved chunks, and Edge-TTS reads it aloud. On question–answer pairs extracted from the user manual of the EC66 cobot (version 6.0.134), the generated answers reached a cosine similarity and Pearson correlation of about 0.95 with the reference answers, and the medium Whisper model achieved 96.9% word accuracy. The evaluation relies on embedding similarity only, uses a single manual in a Chinese-language setup, includes no baseline comparison and does not test unanswerable questions.
Key Insight: Retrieval-augmented answering from a robot manual works even with a small (6B) model.
Relevance: The closest prior work to our project; we add multiple manuals, retrieval baselines and a refusal test.

3.7.4 Hybrid AI Chatbot for Factory Troubleshooting

Authors: K. S. Kiangala and Z. Wang (2024), International Journal of Advanced Manufacturing Technology [28]

The authors build a chatbot human–machine interface for an experimental factory: a LangChain agent connects GPT-3.5 to the factory's monitored data (CSV files), and a Streamlit front end lets operators ask about equipment status and faults in plain English. The chatbot answered 90.9% of 13 troubleshooting queries and all predictive-maintenance queries correctly, in under 10 s on average, and the authors' downtime model estimates that a failure costing 53 min without the chatbot would take about 33 min with it. The chatbot uses structured plant data rather than manuals, the downtime figures are calculated rather than measured, and the authors note GPT-3.5's 4,096-token limit and the API cost.
Key Insight: Natural-language access to plant information can shorten troubleshooting on the shop floor.
Relevance: Links our assistant to the module goal of less downtime and supports a Streamlit interface.

3.7.5 Can a Troubleshooting AI Assistant Improve Task Performance?

Authors: N. Löwhagen, P. Schwendener and T. Netland (2025), International Journal of Production Research [29]

The authors built a multilingual RAG chatbot (LlamaIndex, text-embedding-ada-002, GPT-3.5-turbo) over a train manufacturer's commissioning instructions, maintenance manuals and supplier documents; each answer links to the source document and page. In a controlled field experiment, 19 commissioning technicians solved 173 realistic troubleshooting tasks within 45 minutes each, 15 of them with the assistant and 4 without. Assisted technicians performed significantly better, the gain was largest for less experienced technicians, and the technicians' attitude toward and familiarity with AI moderated the effect. Limitations are the small and unbalanced sample, a controlled rather than live production setting, and the older GPT-3.5 model.
Key Insight: AI assistants help most where experience is lacking, and users value answers that show their sources.
Relevance: Supports the target user of our assistant (operators without a robot expert nearby) and the page-level citations.

3.7.6 HybridRAG for Aircraft Troubleshooting

Authors: X. Xie, X. Tang, S. Gu and L. Cui (2025), Scientific Reports [30]

HybridRAG combines graph-based reasoning over a fault knowledge graph with both vector-based and BM25-based text retrieval, and adds a conversational agent that guides maintenance staff step by step. On a corpus of 1,000 UAV fault-handling records and 20 MB of technical manuals, using the Qwen2-72B model, it improved the F1 score by at least 4% and reduced hallucination by more than 7% compared with mainstream RAG baselines. The authors report failures caused by schema ambiguity and low data quality, and note that building the knowledge graph is costly and that only text data was used.
Key Insight: Combining keyword (BM25), vector and structured retrieval improves fault diagnosis over single-method RAG.
Relevance: Supports a hybrid retrieval design.

3.7.7 Comparative Study of PDF Parsing Tools

Authors: N. S. Adhikari and S. Agarwal (2024), arXiv [31]

The authors compare ten PDF parsing tools, including PyPDF, pdfminer.six, PyMuPDF, pdfplumber, pypdfium2, Unstructured, Tabula, Camelot, Nougat and Table Transformer, on six document categories of the DocLayNet dataset, among them manuals. For text extraction, PyMuPDF and pypdfium generally performed best, and the manual category gave consistently high scores across most tools. For table detection, PyMuPDF performed best on manuals. The manuals in DocLayNet are mainly software manuals and grammar guides rather than industrial machine manuals, and the evaluation of Nougat used a small sample.
Key Insight: Text extraction is a source of error before retrieval even starts.
Relevance: We compare pdfplumber and PyMuPDF on the robot manuals; this study points to PyMuPDF as a strong choice, and Lukens et al. [26] also used it.

3.8 Summary of Reviewed Works

Table 3.1: Summary of the reviewed works.

Ref.	Domain / data	Method	Key result	Use in project
[1]	Manufacturing CPS	5C architecture	Information must reach decision-makers	Positions the assistant
[6, 4, 5]	Cobot safety	Standard, manufacturer guides	Stops need root-cause review	Defines the task
[7]	Maintenance text	Technical language processing	Generic NLP fails on jargon	Mixed question types
[8]	Assistance reports (308 cases)	Fine-tuned Sentence-BERT	MAP@5 0.71–0.76, above BM25	Dense retrieval
[9]	Translation	Transformer	28.4 / 41.0 BLEU	Basis of all models
[10]	General NLP	BERT	GLUE 80.5, SQuAD F1 93.2	Encoder basis
[11]	General LLM	Open-weight models	8B–405B, 128K context	Local LLM
[12]	Text generation	Survey	Retrieved knowledge can reduce hallucination	Plain-LLM baseline
[13]	Text retrieval	BM25	Fast keyword ranking	Keyword baseline
[14]	Open-domain QA	Dense retrieval	+9–19 pts top-20 vs. BM25	Dense retrieval
[15]	Sentence similarity	Sentence-BERT	65 h → 5 s (10k sentences)	Embedding model
[16]	General NLP	Distillation	>99% accuracy at half size	Small encoder

Ref.	Domain / data	Method	Key result	Use in project
[17]	Vector search	FAISS	8.5× faster GPU search	Vector index
[18]	18 IR datasets	Benchmark	BM25 robust across domains	Hybrid search
[19]	TREC, LETOR 3	Rank fusion	Beat Condorcet in all cases; 4–5% better on average	Fusion method
[20]	Open-domain QA	RAG	More specific, factual answers	Core architecture
[21]	RAG for LLMs	Survey	Naive / advanced / modular RAG	Design choices
[22]	Multi-doc QA	Position study	U-shaped accuracy; middle underused	Small top-k
[23]	WikiEval (50 pages)	RAGAS	95% human agreement on faithfulness	Automatic check
[24]	News QA (EN/ZH)	RGB benchmark	Best rejection rate only 45%	Refusal KPI
[25]	Naval CPS manuals	Local RAG	Top-4 hit 41% vs. 19%; 3.6–12.2 s	Citations, latency
[26]	Vehicle fault labels	RAG vs. LLM vs. embedding retrieval	Retrieval 86.3% vs. LLM 69.6%	Baselines
[27]	Cobot manual (EC66)	Voice + RAG	Answer similarity ≈0.95	Closest prior work
[28]	Factory CSV data	GPT-3.5 chatbot	90.9% of 13 queries; est. 53 → 33 min	Industrial context
[29]	Train commissioning	RAG field experiment (19 technicians)	Better scores, largest gain for novices	Target user
[30]	UAV fault records	BM25 + vector + graph	F1 +4%, hallucination −7%	Hybrid idea
[31]	DocLayNet (6 types)	10 PDF parsers	PyMuPDF best for manuals' tables	Parser test

3.9 Synthesis and Observations

The reviewed literature shows five consistent patterns:

- **Grounding beats memory, but only with good retrieval.** Documents make answers more specific and traceable [20, 25, 26], yet an answer is only as good as the passages found.
- **No single retriever wins everywhere.** BM25 is strong for exact terms and new domains [18], dense retrieval for different wording [14, 8], rank fusion combines both [19], and retrieval without an LLM can beat RAG on label matching [26].
- **Wording and preprocessing matter as much as the model.** Imprecise questions more than halved retrieval success in an industrial study [25], the PDF parser changes results [31], and LLMs underuse context in the middle [22].

- **Traceability and refusal are safety requirements.** Robot stops require a root-cause review [5], operators need sources [25, 29], and LLMs are weak at refusing [24].
- **Small local models are feasible with a trade-off.** Local deployment protects company data, while smaller models answer faster but less accurately [11, 25].

3.10 Alignment with the Proposed Work

Table 3.2: Alignment of reviewed research insights with the proposed design.

Criteria	**Research insight**	**Proposed design justification**
Answer reliability	LLMs alone hallucinate [12]	Answer only from manuals with RAG [20]
Operator trust	Operators must cross-check answers [25, 29]	Every answer shows manual and page
Exact codes vs. free text	BM25 for exact terms, dense for paraphrases [14, 18]	Hybrid retrieval fused with RRF [19]
Safety	LLMs are weak at negative rejection [24]	"I don't know" rule and refusal KPI
Fair comparison	Retrieval alone can beat LLMs on fault-label matching [26]	Plain-LLM and BM25 baselines
Context size	Middle context underused [22]	Small top-k, chunk-size tests
Data quality	Parser choice affects results [31]	Compare pdfplumber and PyMuPDF
Privacy and cost	Local models are feasible [11, 25]	Local LLM via Ollama, latency KPI

3.10.1 Identified Research Gaps

- **Few studies on robot error diagnosis.** Industrial RAG studies cover naval systems [25], vehicles [26] and aircraft [30]; the closest robot-specific work [27] uses a single cobot manual and similarity-based evaluation without baselines.
- **Exact codes and free-text symptoms across several vendors' manuals.** Prior work varies the wording of questions within one domain [25] or uses a single robot manual [27]; we found no study that tests exact alphanumeric error codes (e.g., C157A2) together with free-text symptoms across manuals from two vendors (robot and gripper).
- **Hybrid retrieval is rarely compared directly** with keyword and dense retrieval on the same industrial test set.
- **Refusal is rarely measured** in industrial settings, although it is a known weakness of LLMs [24].
- **Low-cost local setups are rarely evaluated**, although they matter for small and medium-sized manufacturers.

4. Method Selection Justification

Based on the literature analysis, the FactoryDocs Assistant is designed as a local, hybrid RAG chatbot over the four robot and gripper manuals. Hybrid retrieval handles both exact error codes and informal problem descriptions, source citations make every answer verifiable, and the refusal rule prevents invented answers in a safety-critical setting.

4.1 Proposed Pipeline

1. **Extraction:** text and tables are extracted from each PDF with pdfplumber, keeping manual name and page; PyMuPDF is tested as an alternative [26, 31].
2. **Chunking:** the text is split into small overlapping chunks tagged with manual and page.
3. **Indexing:** chunks are embedded with a sentence-transformer model (e.g., all-MiniLM-L6-v2) [15, 16] and stored in FAISS [17]; a BM25 index [13] is built over the same chunks.
4. **Retrieval:** the top-k chunks are retrieved by BM25, dense search, or both fused with RRF [19].
5. **Generation:** a local LLM via Ollama (e.g., Llama 3.1 8B, Ollama tag `llama3.1:8b`) [11] answers only from the chunks, cites manual and page, or replies "I don't know".
6. **Interface:** a Streamlit chat app shows the answer with its sources.

4.2 Compared Approaches and Evaluation

All four approaches use the same LLM, prompt structure and 60 questions, so that differences come from retrieval alone (Table 4.1). Chunk sizes (e.g., 400, 800 and 1,000 characters [25, 26]) and top-k values (e.g., 3, 5 and 10) are varied for the best approach. Following the evaluation-first principle [26], the test set and metrics are fixed before tuning (Table 4.2).

Table 4.1: Compared approaches.

#	Approach	Retrieval	Purpose
1	Plain LLM	None	Measures hallucination without documents
2	BM25 RAG	Keyword (BM25)	Baseline for exact codes
3	Dense RAG	Sentence embeddings + FAISS	Handles paraphrased problems
4	Hybrid RAG	BM25 + dense, fused with RRF	Main method, combines both

Table 4.2: Evaluation KPIs and targets.

KPI	Definition	Target
Retrieval hit rate@5	Correct page among the top 5 chunks	$\geq 80\%$
Answer accuracy	Mean rubric score (1 / 0.5 / 0) by two reviewers	$\geq 75\%$, above baselines
Refusal rate	"I don't know" on unanswerable questions	$\geq 80\%$
Response time	Median seconds per answer on a laptop	≤ 10 s

RAGAS faithfulness [23] is computed as an automatic cross-check, false refusals on answerable questions are reported, and results are broken down by question type for error analysis.

4.3 Reasons for Selection

- **RAG:** grounded, traceable and updatable answers without retraining [20].
- **Hybrid retrieval:** exact error codes via BM25 and informal descriptions via dense search [14, 18, 19].
- **Local LLM:** company documents stay on site, at no cost, with acceptable speed [11, 25].
- **Evaluation design:** retrieval, correctness, refusal and speed, as the literature recommends [23, 24, 26].

4.4 Timeline

Table 4.3: Project timeline.

Phase	Date	Deliverable
P1	20 Oct 2026	Proposal, tech stack, MVP definition (revised after feedback)
P2	03 Nov 2026	Bibliography report, PDF extraction, BM25 baseline, test plan
P3	24 Nov 2026	Embeddings and FAISS, first cited answers, plain-LLM baseline
P4	08 Dec 2026	Test set complete, hybrid search, comparison and error analysis
P5	22 Dec 2026	Streamlit demo, demo video, run instructions
Final	12 Jan 2027	IEEE-format report and final presentation

5. Conclusion

The literature review shows that a local, hybrid RAG chatbot is a realistic and well-founded way to speed up robot error diagnosis in production. Unplanned downtime is costly [3], industrial robots are widespread [2], and a protective stop must not be reset before its root cause is reviewed [5]. The knowledge for this review is written in technical language [7], and LLMs used alone can produce unsupported answers [12].

RAG grounds answers in documents and makes them traceable [20, 21]; keyword and dense retrieval have complementary strengths [13, 14, 18]; and RAG must be evaluated on retrieval, answer correctness and refusal [23, 24]. Industrial studies confirm that RAG can support troubleshooting when sources are always shown, that it helps less experienced technicians most, and that wording, PDF extraction and the choice between retrieval and generation matter [25, 26, 29, 31].

Five gaps remain: little work on robot error diagnosis, no study combining exact error codes and free-text symptoms across manuals from several vendors, few direct comparisons of keyword, dense and hybrid retrieval on industrial data, little measurement of refusal, and limited evaluation of low-cost local setups. The FactoryDocs Assistant addresses these gaps with four public UR5e and Robotiq manuals, a 60-question test set in five types, four compared approaches and four KPIs. The next steps are PDF extraction and the BM25 baseline, followed by dense and hybrid retrieval, the full test set, and the comparison and error analysis before the final report on 12 January 2027.

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